

PROJECT PROPOSAL

ISCC Mass Balance Inventory Management System

A web-based system for tracking mass balance bookkeeping under ISCC EU and ISCC PLUS certification

Prepared for	[Client name]
Prepared by	[Your name]
Date	August 4, 2026
Target launch	Week of September 14–21, 2026

1. Executive Summary

This proposal outlines a web application to replace manual, spreadsheet-based mass balance bookkeeping for biomethane and Bio-LNG under ISCC EU and ISCC PLUS certification. The system will track batches, transactions, and balancing periods, and will automatically enforce the compliance rules an ISCC audit checks for — so non-compliant entries are caught at the point of entry rather than discovered later.

The system will be delivered in three phases over approximately six weeks, with a working version covering the core compliance rules ready by the week of September 14–21, 2026. Remaining, lower-priority functionality will follow in a second release shortly after launch.

2. Problem Statement

Mass balance bookkeeping under ISCC involves a number of rules that are easy to get wrong manually and costly to get wrong at audit time, including:

- Credits cannot be pooled across sites without explicit authorization
- Balancing periods must run with no date gaps, and cannot exceed three months
- Every inbound volume requires supporting physical documentation (e.g. a weighbridge ticket)
- Outgoing volume can never exceed what is actually available
- Carry-forward rules differ by scheme — ISCC EU carry-forward is capped by physical stock on hand; ISCC PLUS is not
- Merged batches must carry the worst-case (maximum) GHG value among their sources, not an average
- Booked stock must reconcile with physical stock within a 0.5% tolerance
- Certified volume may move from ISCC EU to ISCC PLUS, but never the reverse

None of these are naturally enforced by a spreadsheet. The goal of this project is to make the compliant path the only path available to the user.

3. Proposed Solution

A web application with three core areas of functionality:

- Data entry: recording inbound, outbound, and conversion transactions, tied to physical documentation where required
- Automatic compliance validation: every entry is checked against the applicable rules in real time, with a clear explanation whenever an entry is rejected
- Auditable period reporting: each balancing period closes with a calculated, traceable balance, and every change made in the system is permanently logged

4. Scope of Work

4.1 Included at Launch (Phase 1 & 2)

Area	Included
Core records	Sites, batches, transactions, physical documents
Compliance rules	Site isolation, period continuity, document requirement, negative-balance prevention, carry-forward capping, GHG max-on-merge, stock tolerance check
Roles	Trader and Processing Unit site types, with conversion tracking for Processing Units
Cross-scheme transfer	One-directional ISCC EU → ISCC PLUS volume transfer
Audit trail	Permanent, append-only log of every change made in the system

4.2 Deferred to Post-Launch (Phase 3)

The following is planned as a follow-on release shortly after launch, and is not required for the system to be considered live and usable:

- A formal material reference list with automated product-group matching (based on density, heating value, and RED category), replacing free-text material entry

5. Technology

The system will be built using widely-supported, production-proven technology:

Layer	Technology
Backend	Node.js (JavaScript/TypeScript) with Express
Database	MySQL 8
Data layer	Prisma ORM
Frontend	React
Hosting	Managed cloud hosting with automated backups

This stack was chosen for reliability and maintainability: a single language (JavaScript/TypeScript) across the full application reduces long-term maintenance overhead, and MySQL provides the data-integrity guarantees the compliance rules depend on.

6. Timeline & Milestones

Phase	Dates	Milestone
Phase 1	Aug 4 – Aug 24, 2026	Core system: records, data entry, foundational compliance rules
Phase 2	Aug 25 – Sep 13, 2026	Carry-forward, stock reconciliation, GHG merge logic, roles, conversions

Phase	Dates	Milestone
Launch	Week of Sep 14 – 21, 2026	System live, covering all rules through cross-scheme transfer
Phase 3	Post-launch	Material reference list and product-group matching

7. Deliverables

- A deployed, working web application, accessible online
- Source code and database schema
- Brief handover documentation covering system setup and how each compliance rule is enforced

8. Assumptions

- Single company, operating under its own ISCC EU and/or ISCC PLUS certification(s) — not a multi-tenant platform for multiple unrelated companies
- Physical stock readings are entered manually; no sensor or IoT integration is included
- No integration with external certification body systems is included in this phase

9. Next Steps

Upon approval of this proposal, work will begin on Phase 1 as outlined above. Progress will be shared at the end of each phase, with the system available for review ahead of the planned launch window.